

# **DAYANANDA SAGAR COLLEGE OF ENGINEERING**

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning**



## **INTERNSHIP REPORT**

**ON**

## **“WEB DEVELOPMENT”**

*Submitted in partial fulfillment for the award of degree(20AI8CINT)*

## **BACHELOR OF ENGINEERING IN**

**Department of Artificial Intelligence and Machine Learning**

*Submitted by:*

**Niranthar M 1DS20AI035**

Conducted at



ISO 9001-2015

(A GOVERNMENT OF KARNATAKA ENTERPRISE)

**KEONICS**

**Department of Artificial Intelligence and Machine Learning  
Dayananda Sagar College of Engineering Bangalore-560078**

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Department of Artificial Intelligence and Machine Learning



## CERTIFICATE

This is to certify that the Internship titled carried out by **Niranthar M [1DS20AI035]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8CINT**)

\_\_\_\_\_  
**Signature of Reporting  
Head**  
**Geetha Jain**

\_\_\_\_\_  
**Signature of Internship  
Coordinator**  
**Prof. Kavya D N**  
Dept. of AI & ML  
DSCE

\_\_\_\_\_  
**Signature of HoD**  
**Dr. Vindhya P Malagi**  
Dept. of AI & ML  
DSCE

### External Viva:

Name of the Examiner

1) \_\_\_\_\_

2) \_\_\_\_\_

Signature with Date

\_\_\_\_\_

\_\_\_\_\_

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning



## D E C L A R A T I O N

I, **Niranthar M**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **KEONICS**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN:1DS20AI035

Place: Bangalore

NAME: NIRANTHAR M

# OFFER LETTER

## SRINIVASA INFOTECH COMPUTERS

1st Floor, Vasudeva Towers, Main Road, BANTWALA-574 211, D.K.  
Ph: 98808 32511 Mob.: 94807 63011, E-mail: geethaiain2010@gmail.com

Franchisee of:



Karnataka State Electronics Development Corporation Limited.  
(Govt. of Karnataka Undertaking)

Regd. Offi. : 2nd floor TTMC 'A' Block-BMTC Shantinagar - K.H. Road - Bangalore - 27

### Free Internship Offer Letter

To,  
Dear Niranthar M  
nirantharnms@gmail.com

As an Associate Website Development Intern, it gives us great pleasure and excitement to welcome you to SHRINIVAS INFOTECH. At Shrinivas infotech, we take pleasure in employing only those individuals who are prepared to make the most of their time and do the best for society. We think that our staff is our greatest strength. We are certain that you will contribute significantly to the venture's overall success. We hope you have the most delightful, educationally rich, and profoundly impactful internship at Shrinivasa infotech. The terms and conditions listed in "Annexure-A" will apply to your appointment. We anticipate having you join us. Please don't be afraid to contact us again if you want further details.

Congratulations!

Sincerely,  
Geetha J Jain  
[Proprietor]

# COMPLETION CERTIFICATE

## SRINIVASA INFOTECH COMPUTERS

1<sup>st</sup> Floor, Vasudeva Towers, Main Road, BANTWALA-574 211, D.K.  
Ph: 98808 32511 Mob.: 94807 63011, E-mail: [geethajain2010@gmail.com](mailto:geethajain2010@gmail.com)

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Karnataka State Electronics Development Corporation Limited.  
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ISO 9001: 2008

Regd. Offi. : 2<sup>nd</sup> floor TTMC 'A' Block-BMTC Shantinagar - K.H. Road - Bangalore - 27

Date: 9 April 2024

To Whom It May Concern,

This letter is to certify that **Niranthar M (1DS20AI035)** student of Dayananda Sagar College of Engineering has successfully completed his internship program of 12 weeks in our organization. His internship period was from 5<sup>th</sup> February 2024 to 8<sup>th</sup> April 2024. He was working with our Technical Department and was actively involved in the tasks assigned to him.

During the span, we found him to be a punctual and reliable person. His learning powers are good, and he picks up quickly. His feedback and evaluation proved that he learned strongly. Moreover, his interpersonal and communication skills are brilliant.

We wish him a bright future.

Sincerely,

Geetha J Jain

  
SRINIVASA INFOTECH COMPUTER  
1<sup>st</sup> Floor, Vasudeva Tower  
Main Road, BANTWALA-574 211 (D.K.)  
(Proprietor),

(SRINIVASA INFOTECH)

## **A C K N O W L E D G E M E N T**

This Internship is a result of accumulated guidance, direction, and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr. Gourav Terwardkar, Filoffee.AI, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-Ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

**NIRANTHAR M [1DS20AI035]**

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# **CHAPTER 1**

## **INTRODUCTION**

During the Internship, I had the opportunity to undertake a web development internship at KEONICS. This experience provided me with invaluable insights into the field of web development, allowing me to apply theoretical knowledge gained from academic studies into practical, real-world projects. Under the guidance of experienced mentors, I delved into various aspects of web development, including front-end and back-end technologies, design principles, and project management methodologies.

Throughout the internship, I actively participated in the development lifecycle of multiple projects, from initial concept and design to implementation and deployment. This report aims to provide a comprehensive overview of my internship experience, highlighting the projects I worked on, the skills I acquired and honed, as well as the challenges faced, and lessons learned along the way.

I am immensely grateful for the opportunity to have been part of the KEONICS. team, and I am excited to share my journey and insights gained during this enriching experience.



## **CHAPTER 2**

### **DESCRIPTION OF ORGANIZATION**

Karnataka State Electronics Development Corporation Limited (KEONICS), an ISO 9001:2008 certified organization, is in existence since 1976 with the sole objective of promoting electronics industries in Karnataka. To ensure that Karnataka is always in the forefront of electronics and IT, we have been providing high quality infrastructure for IT industries, spreading IT education to the masses and providing state-of-the products and service for e-Governance.

KEONICS focuses on setting up world-class infrastructure for the electronics and IT related industries in the state. Aimed at providing a platform for these industries to operate from, we work with Software Technology Parks of India to develop the estates, provide connectivity and administer export-oriented schemes. We have diversified into IT related activities in 1990.

KEONICS' IT solutions span the areas of ERP solutions, web portals, software development, IT consultancy, IT training and networking. Though our major customer base is in Karnataka, we have been awarded with prestigious eGovernance projects including State Wide Area Networking (SWAN) project from Government of Arunachala Pradesh. And so also, our popular web based e-procurement solution is being used by many department of Kerala, Maharastra etc., and Central Public Sectors like HLL, HSCC ltd, Hindustan Shipyard etc., to name a few.

KEONICS has a strong team of professionals possessing adequate skill sets in the areas of networking and IT solutions, providing an efficient and reliable support to customers. In addition to this we are partnered with many industry experts in domain specific verticals for IT product and services area. It is KEONICS' endeavor to provide state-of-the technology solutions to customers to keep pace with the changing technology.

## **CHAPTER 3**

### **SCOPE OF WORK**

During my internship at KEONICS, my primary focus has been on the Web Development aspect within the Developer team. The scope of my work encompasses a range of responsibilities aimed at enhancing the usability, accessibility, and overall user experience of KEONICS's digital products and platforms. Below are the key areas and tasks included in my scope of work:

#### **1. Project Participation:**

- Collaborate with the development team to contribute to the design, development, testing, and deployment of web applications.
- Assist in the creation of project plans, timelines, and deliverables.
- Work on both front-end and back-end development tasks as required by the project.

#### **2. Front-end Development:**

- Utilize HTML, CSS, and JavaScript to build responsive and user-friendly interfaces.
- Implement design mockups provided by the design team using modern front-end frameworks such as React.js or Angular.
- Ensure cross-browser compatibility and optimization for various devices and screen sizes.

#### **3. Back-end Development:**

- Develop server-side logic using programming languages such as Python, Node.js, or PHP.
- Design and implement RESTful APIs to facilitate communication between the front-end and back-end systems.
- Integrate databases (e.g., MySQL, MongoDB) to store and retrieve data efficiently.

#### 4. Version Control and Collaboration:

- Utilize version control systems (e.g., Git) to manage codebase and collaborate with team members.
- Participate in code reviews to ensure code quality, maintainability, and adherence to best practices.

#### 5. Testing and Debugging:

- Write and execute unit tests to validate the functionality of individual components.
- Conduct integration testing to ensure smooth interaction between different modules.
- Identify and troubleshoot bugs and performance issues, providing timely resolutions.

#### 6. Documentation:

- Document technical specifications, project workflows, and codebase architecture.
- Create user manuals or guides for end-users to understand the functionality of the developed applications.

#### 7. Learning and Skill Development:

- Proactively seek opportunities to learn new technologies, frameworks, and best practices in web development.
- Participate in training sessions, workshops, and knowledge-sharing activities organized by the company.

#### 8. Project Management:

- Communicate progress, challenges, and dependencies effectively with project stakeholders.
- Adhere to project timelines and milestones, ensuring timely delivery of assigned tasks.

9. Professional Development:

- Actively participate in performance evaluations and feedback sessions to identify areas for improvement and set professional goals.
- Demonstrate professionalism, teamwork, and a strong work ethic in all aspects of work.

10. Final Report and Presentation:

- Compile a comprehensive internship report documenting the projects undertaken, skills acquired, challenges faced, and lessons learned.
- Deliver a presentation summarizing the internship experience, showcasing key accomplishments and insights gained.

## CHAPTER 4

### LEARNING OBJECTIVES AS GIVEN BY COMPANY

Throughout my internship at KEONICS, the company has outlined specific learning objectives to guide my professional development and maximize the value of the internship experience. These objectives reflect KEONICS's commitment to nurturing talent, fostering growth, and preparing interns for successful careers in the field of IT and software engineering. The following are the key learning objectives provided by KEONICS:

**Mastery of UI/UX Principles:** Gain a deep understanding of user interface (UI) and user experience (UX) design principles, methodologies, and best practices. Learn to apply design thinking techniques to solve complex UI/UX challenges and create intuitive digital experiences for users.

**Technical Proficiency in Design Tools and Technologies:** Acquire proficiency in industry-standard design tools, such as Figma, to create high-fidelity UI mockups and prototypes. Develop skills in API development and Agile and Jira, to implement and iterate on UI designs effectively.

**REST API Development:** Acquire knowledge and proficiency in developing REST API using python, Flask, php and to aid in developing applications.

**User-Centric Design Approach:** Understand the importance of user research and usability testing in informing design decisions and validating design solutions. Learn to empathize with users, identify their needs and pain points, and incorporate user feedback into the design iteration process.

**Collaboration and Communication Skills:** Enhance communication and collaboration skills by effectively articulating design ideas, presenting design concepts, and soliciting feedback from cross-functional teams. Learn to work collaboratively with product managers, developers, and other stakeholders to translate business requirements into user-centric design solutions

**Professional Growth and Adaptability:** Cultivate a growth mindset and a willingness to embrace new challenges, feedback, and opportunities for learning and improvement. Adapt to a dynamic and fast-paced work environment, demonstrating resilience, flexibility, and a proactive approach to problem-solving.

By aligning my learning objectives with the strategic goals and priorities of KEONICS, I aim to leverage this internship opportunity to acquire valuable skills, knowledge, and experiences that will contribute to my professional development and future success in the field of IT and software engineering.

# **CHAPTER 5**

## **CHALLENGES AND SOLUTIONS**

### **Scope Creep**

Challenge: Scope creep arises when additional functionalities or requirements are incorporated into the project beyond the original scope, resulting in delays in the timeline and exceeding allocated resources.

Solution: Employing robust project management strategies like defining clear project scope boundaries, conducting regular scope evaluations, and securing stakeholder approval for any scope modifications. Utilizing Agile methodologies such as Scrum or Kanban can effectively handle changes through incremental development cycles.

### **Content Organization and Prioritization**

Challenge: Deciding the most efficient method to structure and prioritize vast amounts of information in a project, while ensuring completeness and conciseness, can be overwhelming.

Solution: Carrying out comprehensive research to comprehend user needs and usage scenarios, which guides the establishment of a well-organized information structure that highlights critical features and common user activities. Regular feedback sessions with stakeholders aid in refining content organization and ensuring alignment with user expectations.

### **Technology Dependencies and Integration Challenges**

Challenge: Projects frequently depend on external technologies, APIs, or third-party services, resulting in dependencies and integration hurdles that could impact project timelines and functionality.

Solution: Performing thorough research and feasibility studies to identify potential technology dependencies early in the project lifecycle. Establishing transparent communication channels and collaboration agreements with technology providers can help mitigate integration risks, while robust testing and fallback mechanisms can reduce the impact of integration failures.

### **User Adoption and Change Management**

Challenge: Effectively introducing new software or features to end-users and ensuring their acceptance can be difficult, especially when users resist change or are unfamiliar with the new tools.

Solution: Implementing efficient change management tactics, including user training and onboarding programs, communication strategies, and user feedback mechanisms. Involving end-users in the design and development process through user testing and co-creation can enhance support and approval of the new software.



## **CHAPTER 6**

### **ACHIEVEMENTS AND CONTRIBUTION**

Throughout my internship at KEONICS, I had the opportunity to make significant contributions to the development team and the overall success of the project. The following highlights some of the key achievements and contributions made during the internship:

#### **Enhancement of User Experience:**

Through rigorous usability testing and gathering user feedback, I identified pain points and areas for improvement in the user interface and overall user experience. I proposed and implemented iterative design updates and enhancements based on user feedback, resulting in a more intuitive and efficient user experience. I advocated for user-centric design principles and best practices within the development team, fostering a culture of continuous improvement and customer focus.

#### **Cross-Functional Collaboration and Communication:**

I actively engaged in cross-functional collaboration with product managers, designers, developers, and stakeholders to align project goals and priorities effectively. Presenting project updates, findings, and recommendations to senior leadership and stakeholders showcased my professionalism, clarity, and confidence in communication. I played a pivotal role in fostering a culture of teamwork and knowledge sharing within the development team, participating in team meetings, brainstorming sessions, and peer reviews.

#### **Continuous Learning and Professional Development:**

I pursued various opportunities for learning and skill development, including self-directed study, online courses, and participation in company-sponsored training programs. My adaptability and resilience were evident in tackling new challenges and learning new technologies and methodologies relevant to web development. I received positive feedback and recognition from supervisors, colleagues, and stakeholders for my dedication, initiative, and contributions to the success of the project. By leveraging these achievements and contributions, I not only enhanced my skills and knowledge as a web developer but also played a crucial role in the success of the project and the broader objectives of the development team at KEONICS.

## **CHAPTER 7**

### **REFLECTION AND ANALYSIS**

My internship experience at KEONICS has been both enriching and insightful, providing me with valuable opportunities for growth, learning, and self-discovery. As I reflect on my journey and analyze the various aspects of my internship, several key insights and reflections emerge:

#### **Professional Growth and Development:**

Reflecting on my initial expectations and aspirations at the beginning of the internship, I am pleased to acknowledge the significant growth and development I have experienced over the course of my tenure at KEONICS. Through hands-on experience, exposure to real-world projects, and mentorship from seasoned professionals, I have honed my technical skills, expanded my knowledge base, and gained a deeper understanding of Web development. Embracing new challenges, navigating complex projects, and overcoming obstacles have not only enhanced my confidence and competence but also instilled in me a sense of resilience and adaptability crucial for success in the dynamic field of technology.

#### **Project Impact and Contribution:**

Upon reflection, I am gratified by the meaningful contributions I have made to the project and the development team at KEONICS. Developing the web application implementing usability enhancements, and advocating for accessibility and localization initiatives have not only improved the usability and accessibility of the tool but have also fostered a culture of user-centric design and continuous improvement within the team. My contributions have not only demonstrated my technical proficiency and problem-solving skills but have also reinforced the importance of collaboration, communication, and empathy in driving project success.

**Lessons Learned and Areas for Improvement:**

Reflecting on the challenges encountered and the lessons learned during my internship, I recognize the importance of embracing ambiguity, seeking feedback, and learning from failures.

I have learned the value of effective time management, prioritization, and iteration in project execution, as well as the importance of proactive communication and collaboration in cross-functional teams. Moving forward, I am committed to further enhancing my skills, deepening my knowledge, and refining my processes to become a more effective and impactful contributor in future endeavors.

**Personal and Professional Growth:**

Beyond the technical skills and project accomplishments, my internship at KEONICS has also been a journey of personal growth and self-discovery. Engaging with diverse perspectives, collaborating with multidisciplinary teams, and navigating real-world challenges have broadened my horizons, expanded my worldview, and enriched my understanding of the intersection between technology and human experience. I am grateful for the mentorship, support, and camaraderie of my colleagues at KEONICS, whose guidance and encouragement have empowered me to strive for excellence and pursue my passions with purpose and enthusiasm.

## **CHAPTER 8**

### **CONCLUSION**

My internship experience at KEONICS has been nothing short of transformative, providing me with invaluable opportunities for learning, growth, and personal development. Throughout my tenure, I have had the privilege of working alongside dedicated professionals who embody the organization's values of openness, approachability, and fostering a supportive company culture.

During my time at KEONICS, I had the opportunity to contribute to the development of the project, particularly in the creation of the user guide, as well as in enhancing user experience and accessibility. These experiences have not only strengthened my technical skills but have also deepened my understanding of the importance of user-centric design and cross-functional collaboration in delivering impactful solutions.

Moreover, the mentorship and guidance I received from leaders and mentors at KEONICS have been instrumental in shaping my professional journey. Their openness, approachability, and willingness to share their knowledge and insights have been invaluable assets, fostering an environment where I felt supported and empowered to take on new challenges and pursue my passions.

I extend my heartfelt thanks to everyone at KEONICS who has contributed to making my internship experience truly exceptional. I am grateful for the opportunity to have been part of such a dynamic and innovative organization and look forward to applying the lessons learned and making a positive impact in my future endeavors

## **CHAPTER 9**

### **REFERENCES**

Name: Prajeeth

Position: Sr. Developer

Company: KEONICS

Contact: [prajeet\\_j@keonics.com](mailto:prajeet_j@keonics.com)